

# RET fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes RET gene fusions from the msk\_impact\_50k\_2026 study, covering 194 fusion events. 146/194 fusions (75.3%) were in-frame. 144/194 fusions (74.2%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ( $p=9.84123e-05$ , statistically significant at  $\alpha=0.05$ ). Compared to the literature benchmark (PMCID: PMC6430196 (KIF5B-RET): 100.0% in-frame, 100.0% domain-retained), this run observed 75.3% in-frame and 74.2% domain retention.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 194 analyzed fusion events, identifying 33 distinct fusion partner genes. The most frequent partner was KIF5B, observed in 87 events.

### Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 119/146 (81.5%) of in-frame fusions retained the domain,  $p=9.84123e-05$  (statistically significant at  $\alpha=0.05$ ). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Domain-disruption analysis was skipped: RET has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 194 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 1063 aa (permutation-corrected  $p=0.00990099$ , statistically significant at  $\alpha=0.05$ ). This is 58 aa from the nearest configured domain boundary at 1005 aa.

### Corroborating confidence statistics

Corroborating confidence statistics were not computed for this run: Algorithm failed: ValueError: params['group\_field'] is required

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

### Exon retention

Exon retention reported: No target exon was supplied in params or GeneConfig.

## Joint partner

Joint partner reported: Algorithm failed: ValueError: joint\_partner requires a GeneConfig with gene\_pair

# Tables

## composite\_score tables

### composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                    | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------|---------------------|------|
| KIF5B        | 87          | 0.4484536082474227   | 0.20043213737826426    | None                    | 0.6565794315648202       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.40447290221930843 | 1    |
| CCDC6        | 51          | 0.26288659793814434  | 0.20043213737826426    | None                    | 0.7873416786677006       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.3529646650373778  | 2    |
| NCOA4        | 16          | 0.08247422680412371  | 0.20043213737826426    | None                    | 0.8632886064030132       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2919194190660788  | 3    |
| FBXL7        | 2           | 0.010309278350515464 | 0.20043213737826426    | None                    | 1.0                      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.29028688264245817 | 4    |
| CTNNA1       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.9585687382297552       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.27919962404515236 | 5    |
| RUFY2        | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.9510357815442562       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.27758541904111683 | 6    |
| CTNNA3       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.8785310734463276       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.26204869587727503 | 7    |
| RUFY1        | 2           | 0.010309278350515464 | 0.20043213737826426    | None                    | 0.7871939736346516       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2446855912784549  | 8    |
| SPECC1L      | 4           | 0.020618556701030927 | 0.20043213737826426    | None                    | 0.6697269303201507       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2239323441469971  | 9    |
| ERCC6        | 3           | 0.015463917525773196 | 0.20043213737826426    | None                    | 0.6704331450094162       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.22187454479101493 | 10   |
| TIMM23B      | 3           | 0.015463917525773196 | 0.20043213737826426    | None                    | 0.6701192718141871       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.22180728624918014 | 11   |
| RUFY3        | 2           | 0.010309278350515464 | 0.20043213737826426    | None                    | 0.6694915254237288       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2194636380904004  | 12   |
| CLIP1        | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6704331450094162       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21745628264079403 | 13   |
| GRIPAP1      | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6704331450094162       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21745628264079403 | 14   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                    | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------|---------------------|------|
| RASSF4       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6704331450094162       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21745628264079403 | 15   |
| SHROOM3      | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6704331450094162       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21745628264079403 | 16   |
| TRIM24       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6704331450094162       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.21745628264079403 | 17   |
| CCDC186      | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6694915254237288       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2172545070152896  | 18   |
| CUBN         | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6694915254237288       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2172545070152896  | 19   |
| GEMIN5       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6694915254237288       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2172545070152896  | 20   |
| RELCH        | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6694915254237288       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2172545070152896  | 21   |
| SLC4A4       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6694915254237288       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.2172545070152896  | 22   |
| EML4         | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6327683615819208       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.20938525762061647 | 23   |
| RBPMS        | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.6177024482109228       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.20615684761254544 | 24   |
| ERC1         | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.60075329566855         | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.20252488635346555 | 25   |
| PDCD10       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.5894538606403013       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.20010357884741226 | 26   |
| TFG          | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.5894538606403013       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.20010357884741226 | 27   |
| LIN52        | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.551789077212806        | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.19203255382723472 | 28   |
| DOCK1        | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.5225988700564972       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.1857775094365971  | 29   |
| KCNMA1       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.5169491525423728       | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.18456685568357045 | 30   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                    | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|----------------------------------------------------------|---------------------|------|
| KIAA1217     | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.39924670433145004      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.15934490249551556 | 31   |
| UXS1         | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.20527306967984937      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.11777912364160113 | 32   |
| KIF13A       | 1           | 0.005154639175257732 | 0.20043213737826426    | None                    | 0.0                      | None                       | ['cutpoint_proximity', 'domain_retention', 'recurrence'] | 0.07379203728163342 | 33   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value                | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|---------------------|------------------------|---------------------|
| 1        | 1                      | 143              | 1                      | 49               | 0.34265734265734266 | 0.45002937877250143    | 0.34675913373064243 |
| 219      | 1                      | 143              | 2                      | 48               | 0.16783216783216784 | 0.16358634688317933    | 0.7862529459014297  |
| 425      | 2                      | 142              | 2                      | 48               | 0.3380281690140845  | 0.2738144245997509     | 0.5625436768774693  |
| 550      | 4                      | 140              | 3                      | 47               | 0.44761904761904764 | 0.37669413920559647    | 0.424011136713296   |
| 556      | 5                      | 139              | 3                      | 47               | 0.5635491606714629  | 0.4269916226168348     | 0.369580645553364   |
| 587      | 6                      | 138              | 4                      | 46               | 0.5                 | 0.28474795225204086    | 0.5455393904405627  |
| 622      | 7                      | 137              | 4                      | 46               | 0.5875912408759124  | 0.47853605021856643    | 0.32008533934743655 |
| 627      | 8                      | 136              | 6                      | 44               | 0.43137254901960786 | 0.19970190836413992    | 0.699617784963772   |
| 639      | 9                      | 135              | 6                      | 44               | 0.4888888888888889  | 0.22050970732209507    | 0.6565722871774424  |
| 657      | 10                     | 134              | 6                      | 44               | 0.5472636815920398  | 0.3689676566450757     | 0.43301170201493716 |
| 663      | 11                     | 133              | 6                      | 44               | 0.606516290726817   | 0.3861986635634364     | 0.4131892334256796  |
| 671      | 13                     | 131              | 6                      | 44               | 0.727735368956743   | 0.5831276940464237     | 0.2342363324669072  |
| 673      | 14                     | 130              | 6                      | 44               | 0.7897435897435897  | 0.6010603140306741     | 0.22108194607309356 |
| 687      | 15                     | 129              | 6                      | 44               | 0.8527131782945736  | 0.7931139769308365     | 0.1006643965471982  |
| 708      | 16                     | 128              | 6                      | 44               | 0.9166666666666666  | 0.8019071605022902     | 0.09587590854306939 |
| 712      | 74                     | 70               | 8                      | 42               | 5.55                | 1.0784500183907644e-05 | 4.967199977812329   |
| 713      | 141                    | 3                | 14                     | 36               | 120.85714285714286  | 3.386108118610997e-24  | 23.470299178959955  |
| 934      | 141                    | 3                | 15                     | 35               | 109.66666666666667  | 3.253299303362719e-23  | 22.487675979810888  |
| 1019     | 142                    | 2                | 15                     | 35               | 165.66666666666666  | 2.831240208694676e-24  | 23.548023282565293  |
| 1063     | 144                    | 0                | 15                     | 35               | None                | 5.1561797693606186e-27 | 26.287671949448907  |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|     |    |
|-----|----|
| 119 | 25 |
| 27  | 23 |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| CCDC186      | 1           |
| CCDC6        | 51          |
| CLIP1        | 1           |
| CTNNA1       | 1           |
| CTNNA3       | 1           |
| CUBN         | 1           |
| DOCK1        | 1           |
| EML4         | 1           |
| ERC1         | 1           |
| ERCC6        | 3           |
| FBXL7        | 2           |
| GEMIN5       | 1           |
| GRIPAP1      | 1           |
| KCNMA1       | 1           |
| KIAA1217     | 1           |
| KIF13A       | 1           |
| KIF5B        | 87          |
| LIN52        | 1           |
| NCOA4        | 16          |
| PDCD10       | 1           |
| RASSF4       | 1           |
| RBPMS        | 1           |
| RELCH        | 1           |
| RUFY1        | 2           |
| RUFY2        | 1           |
| RUFY3        | 2           |
| SHROOM3      | 1           |
| SLC4A4       | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| SPECC1L      | 4           |
| TFG          | 1           |
| TIMM23B      | 3           |
| TRIM24       | 1           |
| UXS1         | 1           |

## Per-event results (results.tsv)

| event_id                    | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains | lost_domains          | disrupted_domains | source_annotation_text                                                                                                 | source_site2_effect_on_frame |
|-----------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|------------------|-----------------------|-------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00517<br>81-T03-IM6-3 | P-00517<br>81-T03-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | thre_prime  | 61651178           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-00598<br>57-T02-IM7-4 | P-00598<br>57-T02-IM7 |            | CCDC6-RET   | CCDC6        | in-frame     | thre_prime  | 61604308           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.453+8003:CCDC6_c.2137-417:RETinv   Protein Fusion: in frame {CCDC6:RET}  | NA                           |
| EVT-P-00659<br>92-T01-IM7-5 | P-00659<br>92-T01-IM7 |            | CCDC6-RET   | CCDC6        | in-frame     | thre_prime  | 61653659           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12221:CCDC6_c.2137-334:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-00494<br>61-T02-IM6-6 | P-00494<br>61-T02-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | thre_prime  | 61615910           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-3450:CCDC6_c.2137-327:RETinv   Protein Fusion: in frame {CCDC6:RET}  | NA                           |
| EVT-P-00363<br>79-T01-IM6-7 | P-00363<br>79-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | thre_prime  | 61639380           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26500:CCDC6_c.2137-614:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-00374<br>12-T01-IM6-8 | P-00374<br>12-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | thre_prime  | 61624434           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-11974:CCDC6_c.2136+655:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-00486<br>11-T01-IM6-9 | P-00486<br>11-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | thre_prime  | 61652942           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+12938:CCDC6_c.2137-795:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                         | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|------------------|-----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00648<br>53-T01-IM7-10 | P-00648<br>53-T01-IM7 |            | CCDC6-RET   | CCDC6        | out-of-frame | three_prime | 61629044           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-16584:CCDC6_c.1879+388:RETinv   Protein Fusion: out of frame {CCDC6:RET}                                     | NA                           |
| EVT-P-00471<br>44-T01-IM6-11 | P-00471<br>44-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61657033           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8847:CCDC6_c.2137-144:RETinv   Protein Fusion: in frame {CCDC6:RET}                                          | NA                           |
| EVT-P-00382<br>57-T01-IM6-12 | P-00382<br>57-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61614591           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6(NM_005436) - RET (NM_020975.4) Fusion (CCDC6 exon1 fused with RET exon12):c.304-2131:CCDC6_c.2136+410:RETinv   Protein Fusion: in frame {CCDC6:RET}      | NA                           |
| EVT-P-00173<br>65-T04-IM6-13 | P-00173<br>65-T04-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61640379           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-00395<br>48-T01-IM6-14 | P-00395<br>48-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61621961           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-9501:CCDC6_c.2136+659:RETinv   Protein Fusion: in frame {CCDC6:RET}                                          | NA                           |
| EVT-P-00401<br>11-T01-IM6-15 | P-00401<br>11-T01-IM6 |            | CCDC6-RET   | CCDC6        | unknown      | three_prime | 61626333           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-13873:CCDC6_c.1919:RETinv   Protein Fusion: mid-exon {CCDC6:RET}                                             | NA                           |
| EVT-P-00411<br>92-T01-IM6-16 | P-00411<br>92-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61657248           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+8632:CCDC6_c.2136+387:RETinv   Protein Fusion: in frame {CCDC6:RET}                                          | NA                           |
| EVT-P-00226<br>42-T04-IM6-17 | P-00226<br>42-T04-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61626673           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv   Protein Fusion: in frame {CCDC6:RET}                                         | NA                           |
| EVT-P-00226<br>42-T03-IM6-18 | P-00226<br>42-T03-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61626673           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-14213:CCDC6_c.2137-264:RETinv   Protein Fusion: in frame {CCDC6:RET}                                         | NA                           |
| EVT-P-00517<br>81-T01-IM6-19 | P-00517<br>81-T01-IM6 |            | CCDC6-RET   | CCDC6        | in-frame     | three_prime | 61651178           | 20              | 1115                        | True                   | lost          |                  | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+14702:CCDC6_c.2136+193:RETinv   Protein Fusion: in frame {CCDC6:RET}                                         | NA                           |

| event_id                     | sample_id             | patient_id | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00424<br>17-T01-IM6-20 | P-00424<br>17-T01-IM6 |            | CCDC6-RET    | CCDC6        | in-frame     | thre_prime  | 61649492           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+16388 :CCDC6_c.2136+213:RETinv   Protein Fusion: in frame {CCDC6:RET}                                              | NA                           |
| EVT-P-00049<br>01-T01-IM5-21 | P-00049<br>01-T01-IM5 |            | CCDC6-RET    | CCDC6        | in-frame     | thre_prime  | 61653032           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) reciprocal fusion (CCDC6 exon1 with RET exons 12-20) : :CCDC6_c.2136+213:RETinv   Protein fusion: in frame (CCDC6-RET)           | NA                           |
| EVT-P-00223<br>80-T03-IM6-22 | P-00223<br>80-T03-IM6 |            | CCDC6-RET    | CCDC6        | in-frame     | thre_prime  | 61644070           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+21810 :CCDC6_c.2136+475:RETinv   Protein Fusion: in frame {CCDC6:RET}                                              | NA                           |
| EVT-P-00685<br>00-T01-IM7-23 | P-00685<br>00-T01-IM7 |            | CCDC6-RET    | CCDC6        | in-frame     | thre_prime  | 61654590           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+11290 :CCDC6_c.2136+138:RETinv   Protein Fusion: in frame {CCDC6:RET}                                              | NA                           |
| EVT-P-00577<br>35-T01-IM6-24 | P-00577<br>35-T01-IM6 |            | CCDC6-RET    | CCDC6        | in-frame     | thre_prime  | 61665755           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+125:CCDC6_c.2137-145:RETinv   Protein Fusion: in frame {CCDC6:RET}                                                 | NA                           |
| EVT-P-00380<br>53-T01-IM6-26 | P-00380<br>53-T01-IM6 |            | CLIP1-RET    | CLIP1        | unknown      | thre_prime  | 43611121           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CLIP1 (NM_001247997) - RET (NM_020975) rearrangement: t(10;12)(q11.21;q24.31)(chr10:g.43611121::chr12:g.122825512)   Protein Fusion: mid-exon {CLIP1:RET}            | NA                           |
| EVT-P-00002<br>52-T01-IM3-27 | P-00002<br>52-T01-IM3 |            | CUBN-RET     | CUBN         | unknown      | thre_prime  | 43610494           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | NA   Antisense fusion                                                                                                                                                | NA                           |
| EVT-P-00382<br>56-T01-IM6-28 | P-00382<br>56-T01-IM6 |            | ERC1-RET     | ERC1         | unknown      | thre_prime  | 43609964           | 11              | 639                         | False                  | retained      | Protein kinase domain |                       |                   | ERC1 (NM_178040) - RET (NM_020975) Rearrangement : t(10;12)(q11.1; p13.32)(chr10:g.43609964::chr12:g.1264106)   Protein Fusion: mid-exon {ERC1:RET}                  | NA                           |
| EVT-P-00347<br>81-T01-IM6-29 | P-00347<br>81-T01-IM6 |            | GRIPAP1-RET  | GRIPAP1      | in-frame     | thre_prime  | 43611853           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | GRIPAP1 (NM_020137) - RET (NM_020975) rearrangement: t(X;10)(p11.23q11.21)(chrX:g.48831008::chr10:g.43611853)   Protein Fusion: in frame {GRIPAP1:RET}               | NA                           |
| EVT-P-00262<br>45-T01-IM6-30 | P-00262<br>45-T01-IM6 |            | KIAA1217-RET | KIAA1217     | unknown      | thre_prime  | 43606666           | 7               | 425                         | False                  | retained      | Protein kinase domain |                       |                   | KIAA1217 (NM_019590) - RET (NM_020975) fusion (KIAA1217 exons 1-13 fused to RET exons 7-20): c.2390:KIAA1217_c.1275:RETdel   Protein Fusion: mid-exon {KIAA1217:RET} | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role  | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                                                                                                                            | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|--------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0042153-T01-IM6-31 | P-0042153-T01-IM6 |            | KIF13A-RET  | KIF13A       | in-frame     | thre_e_prime | 17806558           | 1               | 1                           | True                   | retained      | Protein kinase domain |              |                   | KIF13A (NM_022113) - RET (NM_020975) fusion: t(6;10)(p22.3;q11.21)(chr6:g.17806558::chr10:g.43610510)   Protein Fusion: in frame {KIF13A:RET}                                                                                                                                                     | NA                           |
| EVT-P-0019535-T02-IM6-32 | P-0019535-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611913           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-2113:KIF5B_c.2137-119:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                                                                                | NA                           |
| EVT-P-0039257-T03-IM7-33 | P-0039257-T03-IM7 |            | KIF5B-RET   | KIF5B        | unknown      | thre_e_prime | 43610060           | 11              | 671                         | False                  | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv   Protein Fusion: mid-exon {KIF5B:RET}                                                                                                                                                                                | NA                           |
| EVT-P-0009514-T01-IM5-34 | P-0009514-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611811           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | Note: The KIF5B (NM_004521) - RET (NM_020975) reciprocal rearrangement event is an inversion which results in the fusion of KIF5B exons 1 to 15 and RET exons 12 to 20. The clinical trial research is currently ongoing with cabozantinib. PMID: 23533264   Protein fusion: in frame (KIF5B-RET) | NA                           |
| EVT-P-0055183-T01-IM6-35 | P-0055183-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43610308           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                                                                                                                            | NA                           |
| EVT-P-0011035-T01-IM5-36 | P-0011035-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611322           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20): c.1726-2296:KIF5B_c.2137-710:RETinv   Protein fusion: in frame (KIF5B-RET)                                                                                                                              | NA                           |
| EVT-P-0012648-T01-IM5-37 | P-0012648-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611697           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-335:RETinv   Protein fusion: in frame (KIF5B-RET)                                                                                                                                                                            | NA                           |
| EVT-P-0053175-T01-IM6-38 | P-0053175-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611162           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-282:KIF5B_c.2137-870:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                                                                                                                             | NA                           |
| EVT-P-0052861-T02-IM6-39 | P-0052861-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43610260           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-20:KIF5B_c.2136+76:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                                                                                                                               | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0013825-T01-IM5-40 | P-0013825-T01-IM5 |            | KIF5B-RET   | KIF5B        | unknown      | three_prime | 43611520           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1782:KIF5B_c.2137-512:RETinv   Protein fusion: mid-exon (KIF5B-RET)      | NA                           |
| EVT-P-0052826-T01-IM6-41 | P-0052826-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611182           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2358:KIF5B_c.2137-850:RETinv   Protein Fusion: in frame (KIF5B:RET)                                             | NA                           |
| EVT-P-0014808-T01-IM6-42 | P-0014808-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611055           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv   Protein fusion: in frame (KIF5B-RET) | NA                           |
| EVT-P-0034357-T01-IM6-43 | P-0034357-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611187           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1863:KIF5B_c.2137-845:RETinv   Protein Fusion: in frame (KIF5B:RET)                                             | NA                           |
| EVT-P-0014808-T02-IM6-44 | P-0014808-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611056           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1765:KIF5B_c.2136+871:RETinv   Protein Fusion: in frame (KIF5B:RET) | NA                           |
| EVT-P-0033700-T02-IM6-45 | P-0033700-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611670           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1454:KIF5B_c.2137-362:RET   Protein Fusion: in frame (KIF5B:RET)                                                | NA                           |
| EVT-P-0014808-T03-IM6-46 | P-0014808-T03-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611056           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1764:KIF5B_c.2136+872:RETinv   Protein Fusion: in frame (KIF5B:RET)                                             | NA                           |
| EVT-P-0008289-T01-IM5-47 | P-0008289-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611748           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20) : c.1726-824:KIF5B_c.2137-284:RETinv   Protein fusion: in frame (KIF5B-RET) | NA                           |
| EVT-P-0000252-T01-IM3-48 | P-0000252-T01-IM3 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610597           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | RET (NM_020975) Inversion (c.2136+413_KIF5B:c.1726-153inv)   Protein fusion: in frame (KIF5B-RET)                                                                  | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0008101-T01-IM5-49 | P-0008101-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610455           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-18 fused with RET exons 12-20) : c.2095-38:KIF5B_c.2136+271:RETinv   Protein fusion: in frame {KIF5B:RET}         | NA                           |
| EVT-P-0048968-T01-IM6-50 | P-0048968-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611809           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+211:KIF5B_c.2137-223:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                       | NA                           |
| EVT-P-0048829-T01-IM6-51 | P-0048829-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610616           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+948:KIF5B_c.2136+432:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                       | NA                           |
| EVT-P-0065876-T01-IM7-52 | P-0065876-T01-IM7 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611299           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1902:KIF5B_c.2137-733:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                      | NA                           |
| EVT-P-0016221-T01-IM6-53 | P-0016221-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611268           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused in-frame to RET exons 12-20) : c.1725+275:KIF5B_c.2137-764:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-0037367-T01-IM6-54 | P-0037367-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611874           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1038:KIF5B_c.2137-158:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                      | NA                           |
| EVT-P-0057072-T02-IM6-55 | P-0057072-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611106           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508:KIF5B_c.2136+922:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                      | NA                           |
| EVT-P-0017214-T01-IM6-56 | P-0017214-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610647           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused in-frame with RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-0017214-T02-IM6-57 | P-0017214-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610647           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-23 fused to RET exons 12-20): c.2545-27:KIF5B_c.2136+463:RETinv   Protein Fusion: in frame {KIF5B:RET}            | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role  | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|--------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0032753-T01-IM6-58 | P-0032753-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43610925           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1323 :KIF5B_c.2136+741:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                           |
| EVT-P-0057072-T03-IM7-59 | P-0057072-T03-IM7 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611106           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2508 :KIF5B_c.2136+922:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                           |
| EVT-P-0032686-T01-IM6-60 | P-0032686-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611982           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2299 :KIF5B_c.2137-50:RETinv   Protein Fusion: in frame {KIF5B:RET}                                               | NA                           |
| EVT-P-0017523-T02-IM6-61 | P-0017523-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611192           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion ( KIF5B exons 1-15 fused to RET exons 12-20): c. 1726-2362:KIF5B_c.2137-840:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-0046827-T02-IM6-62 | P-0046827-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43610889           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1305: KIF5B_c.2136+705:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                           |
| EVT-P-0032407-T01-IM6-63 | P-0032407-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43611730           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1914+230: KIF5B_c.2137-302:RETinv   Protein Fusion: in frame {KIF5B:RET}                                               | NA                           |
| EVT-P-0018069-T01-IM6-64 | P-0018069-T01-IM6 |            | KIF5B-RET   | KIF5B        | out-of-frame | five_prime   | 43607832           | 8               | 550                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.1648+160:RET_c.1915-72:KIF5Binv   Protein Fusion: out of frame {RET:KIF5B}                                      | NA                           |
| EVT-P-0005573-T01-IM5-65 | P-0005573-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43610418           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c. 1726-860:KIF5B_c.2136+234:RETinv   Protein fusion: in frame (KIF5B-RET)          | NA                           |
| EVT-P-0004995-T01-IM5-66 | P-0004995-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_e_prime | 43610660           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B(NM_004521) -RET (NM_020975) Fusion ( KIF5B exon 15 fused to RET exon 12) : c.1726-2366:KIF5B_c.2136+476:RETinv   Protein fusion: in frame (KIF5B-RET)          | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                                            | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0036092-T01-IM6-67 | P-0036092-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610413           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1138:KIF5B_c.2136+229:RETinv   Protein Fusion: in frame {KIF5B:RET}                                | NA                           |
| EVT-P-0046726-T01-IM6-68 | P-0046726-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610987           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+768:KIF5B_c.2136+803:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                             | NA                           |
| EVT-P-0032227-T01-IM6-69 | P-0032227-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611125           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2000:KIF5B_c.2137-907:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                            | NA                           |
| EVT-P-0020273-T02-IM6-70 | P-0020273-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611819           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1914+131:KIF5B_c.2137-213:RETinv   Protein Fusion: in frame {KIF5B:RET}                                 | NA                           |
| EVT-P-0043007-T02-IM6-71 | P-0043007-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610813           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                                             | NA                           |
| EVT-P-0021150-T01-IM6-72 | P-0021150-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611591           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+312:KIF5B_c.2137-441:RETinv   Protein Fusion: in frame {KIF5B:RET}                                 | NA                           |
| EVT-P-0021578-T01-IM6-73 | P-0021578-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610809           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1231:KIF5B_c.2136+624:RETinv   Protein Fusion: in frame {KIF5B:RET}                                | NA                           |
| EVT-P-0021879-T01-IM6-74 | P-0021879-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610699           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B1 exons 1-15 fused with RET exons 12-20 including the kinase domain.): c.1726-882:KIF5B_c.2136+515_RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-0021937-T01-IM6-75 | P-0021937-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611555           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20): c.1725+886:KIF5B_c.2137-477:RETinv   Protein Fusion: in frame {KIF5B:RET}                               | NA                           |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                 | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00219<br>37-T03-IM6-76 | P-00219<br>37-T03-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611555           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+886:KIF5B_c.2137-477:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                           |
| EVT-P-00430<br>07-T01-IM6-77 | P-00430<br>07-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610813           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.2544+168:KIF5B_c.2136+629:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                           |
| EVT-P-00426<br>08-T01-IM6-78 | P-00426<br>08-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611802           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1670:KIF5B_c.2137-230:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-00425<br>80-T01-IM6-79 | P-00425<br>80-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610944           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1619:KIF5B_c.2136+760:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-00599<br>13-T01-IM7-80 | P-00599<br>13-T01-IM7 |            | KIF5B-RET   | KIF5B        | unknown      | thre_prime  | 43610108           | 11              | 687                         | False                  | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2441:KIF5B_c.2060:RETinv   Protein Fusion: mid-exon {KIF5B:RET}     | NA                           |
| EVT-P-00010<br>10-T02-IM5-81 | P-00010<br>10-T02-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610863           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-780:KIF5B_c.2136+679:RETinv   Protein fusion: in frame (KIF5B-RET)  | NA                           |
| EVT-P-00607<br>48-T02-IM7-82 | P-00607<br>48-T02-IM7 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611527           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-00421<br>04-T01-IM6-83 | P-00421<br>04-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610505           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-1056:KIF5B_c.2136+321:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-00031<br>25-T01-IM5-84 | P-00031<br>25-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611584           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | NA   Protein fusion: in frame (KIF5B-RET)                                                                              | NA                           |
| EVT-P-00031<br>25-T01-IM5-85 | P-00031<br>25-T01-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | five_prime  | 43611581           | 12              | 713                         | True                   | lost          |                       | Protein kinase domain |                   | NA   Protein fusion: in frame (RET-KIF5B)                                                                              | NA                           |
| EVT-P-00028<br>73-T01-IM3-86 | P-00028<br>73-T01-IM3 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611244           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | NA   Protein fusion: in frame (KIF5B-RET)                                                                              | NA                           |

| event_id                 | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0022866-T01-IM6-87 | P-0022866-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611871           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-767:KIF5B_c.2137-161:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                           |
| EVT-P-0024225-T01-IM6-88 | P-0024225-T01-IM6 |            | KIF5B-RET   | KIF5B        | unknown      | thre_prime  | 43609110           | 10              | 622                         | False                  | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 10-20): c.2762-430:KIF5B_c.1866:RETinv   Protein Fusion: mid-exon {KIF5B:RET}      | NA                           |
| EVT-P-0002694-T02-IM5-89 | P-0002694-T02-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611682           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | NA   Protein fusion: in frame (KIF5B-RET)                                                                                                                          | NA                           |
| EVT-P-0024843-T01-IM6-90 | P-0024843-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611733           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20): c.1726-1339:KIF5B_c.2137-299:RETinv   Protein Fusion: in frame {KIF5B:RET}     | NA                           |
| EVT-P-0025100-T01-IM6-91 | P-0025100-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611511           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-2273:KIF5B_c.2137-521:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-0025100-T02-IM6-92 | P-0025100-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611511           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) and RET (NM_020975) Fusion (KIF5B exon 16 fused to RET exon 12): c.1726-2273:KIF5B_c.2137-788:RETinv   Protein Fusion: in frame {KIF5B:RET}      | NA                           |
| EVT-P-0039941-T01-IM6-93 | P-0039941-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611339           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+2040:KIF5B_c.2137-693:RETinv   Protein Fusion: in frame {KIF5B:RET}                                             | NA                           |
| EVT-P-0060748-T03-IM7-94 | P-0060748-T03-IM7 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611527           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2304:KIF5B_c.2137-505:RETinv   Protein Fusion: in frame {KIF5B:RET}                                             | NA                           |
| EVT-P-0039474-T02-IM6-95 | P-0039474-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610359           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                           |
| EVT-P-0039474-T01-IM6-96 | P-0039474-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610359           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-203:KIF5B_c.2136+175:RETinv   Protein Fusion: in frame {KIF5B:RET}                                              | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0055183-T02-IM6-97  | P-0055183-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610308           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1798:KIF5B_c.2136+124:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                           |
| EVT-P-0025838-T01-IM6-98  | P-0025838-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610972           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion ( KIF5B exons 1-15 fused with RET exons 12-20.): c.1726-601:KIF5B_c.2136+788:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-0025928-T01-IM6-99  | P-0025928-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611910           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+483:KIF5B_c.2137-122C>T:RETinv   Protein Fusion: in frame {KIF5B:RET}  | NA                           |
| EVT-P-0025928-T02-IM6-100 | P-0025928-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611910           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+483:KIF5B_c.2137-122:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                 | NA                           |
| EVT-P-0036637-T01-IM6-101 | P-0036637-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43611965           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1726-1731:KIF5B_c.2137-67:RETinv   Protein Fusion: in frame {KIF5B:RET}     | NA                           |
| EVT-P-0039257-T01-IM6-102 | P-0039257-T01-IM6 |            | KIF5B-RET   | KIF5B        | unknown      | thre_prime  | 43610060           | 11              | 671                         | False                  | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1132:KIF5B_c.2012:RETinv   Protein Fusion: mid-exon {KIF5B:RET}                                                    | NA                           |
| EVT-P-0026693-T01-IM6-103 | P-0026693-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610511           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion:c.2544+156:KIF5B_c.2136+327:RETinv.   Protein Fusion: in frame {KIF5B:RET}                                                 | NA                           |
| EVT-P-0026777-T01-IM6-104 | P-0026777-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610717           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv   Protein Fusion: in frame {KIF5B:RET}    | NA                           |
| EVT-P-0026777-T02-IM6-105 | P-0026777-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610717           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-16 fused to RET exons 12-20): c.1726-1554:KIF5B_c.2136+533:RETinv   Protein Fusion: in frame {KIF5B:RET}    | NA                           |
| EVT-P-0038961-T01-IM6-106 | P-0038961-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | thre_prime  | 43610775           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-293:KIF5B_c.2136+591:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                 | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                               | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00277<br>23-T01-IM6-107 | P-00277<br>23-T01-IM6 |            | KIF5B-RET   | KIF5B        | unknown      | three_prime | 43610171           | 11              | 708                         | False                  | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 11-20): c.1725+966:KIF5B_c.2123:RETinv   Protein Fusion: mid-exon {KIF5B:RET}        | NA                           |
| EVT-P-00367<br>37-T02-IM6-108 | P-00367<br>37-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43612025           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused to RET exons 12-20): c.1725+1487:KIF5B_c.2137-7:RETinv   Protein Fusion: in frame {KIF5B:RET}     | NA                           |
| EVT-P-00613<br>31-T01-IM7-109 | P-00613<br>31-T01-IM7 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610884           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1915-286:KIF5B_c.2136+700:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                           |
| EVT-P-00279<br>91-T01-IM6-110 | P-00279<br>91-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43608201           | 9               | 550                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-24 fused to RET exons 9-20): c.2762-248:KIF5B_c.1649-100:RETinv   Protein Fusion: in frame {KIF5B:RET}     | NA                           |
| EVT-P-00629<br>48-T03-IM7-111 | P-00629<br>48-T03-IM7 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610211           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1725+1625:KIF5B_c.2136+27:RETinv   Protein Fusion: in frame {KIF5B:RET}                                                | NA                           |
| EVT-P-00649<br>51-T01-IM7-112 | P-00649<br>51-T01-IM7 |            | KIF5B-RET   | KIF5B        | out-of-frame | three_prime | 43611130           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.2204+2:KIF5B_c.2137-902:RETinv   Protein Fusion: out of frame {KIF5B:RET}                                              | NA                           |
| EVT-P-00304<br>53-T01-IM6-113 | P-00304<br>53-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43610911           | 11              | 712                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion: c.1726-2396:KIF5B_c.2136+727:RETinv   Protein Fusion: in frame {KIF5B:RET}                                               | NA                           |
| EVT-P-00374<br>94-T01-IM6-114 | P-00374<br>94-T01-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611755           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) Fusion (KIF5B exon15 fused with RET exon12): c.1725+985:KIF5B_c.2137-277:RETinv   Protein Fusion: in frame {KIF5B:RET}           | NA                           |
| EVT-P-00002<br>08-T02-IM5-115 | P-00002<br>08-T02-IM5 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611647           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 fused with RET exons 12-20): c.1726-1971:KIF5B_c.2137-385:RETinv   Protein fusion: in frame (KIF5B-RET) | NA                           |
| EVT-P-00002<br>08-T01-IM3-116 | P-00002<br>08-T01-IM3 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611647           | 12              | 713                         | True                   | retained      | Protein kinase domain |              |                   | RET (NM_020975) Inversion (c.2137-385_KIF5B:c.1726-1971inv)   Protein fusion: in frame (KIF5B-RET)                                                                   | NA                           |

| event_id                          | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                                    | source_site2_effect_on_frame |
|-----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00162<br>21-T02-IM6-1<br>17 | P-00162<br>21-T02-IM6 |            | KIF5B-RET   | KIF5B        | in-frame     | three_prime | 43611268           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | KIF5B (NM_004521) - RET (NM_020975) fusion (KIF5B exons 1-15 with RET exons 12-20 including the kinase domain): c.1725+275:KIF5B_c.2137-764:RETinv   Protein Fusion: in frame {KIF5B:RET} | NA                           |
| EVT-P-00196<br>63-T01-IM6-1<br>18 | P-00196<br>63-T01-IM6 |            | LIN52-RET   | LIN52        | out-of-frame | five_prime  | 43608613           | 9               | 587                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - LIN52 (NM_001024674) Rearrangement: t(10;14)(p32.1;q12)(chr10:g.43608613:chr14:g.74562750)   Protein Fusion: out of frame {RET:LIN52}                                   | NA                           |
| EVT-P-00658<br>26-T01-IM7-1<br>20 | P-00658<br>26-T01-IM7 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51587635           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2136+673:RET_c.*23+937NCOA4dup   Protein Fusion: out of frame {TIMM23B:RET}                                                              | NA                           |
| EVT-P-00379<br>61-T01-IM6-1<br>21 | P-00379<br>61-T01-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51584256           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.763-360:NCOA4_c.2137-584:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                      | NA                           |
| EVT-P-00383<br>15-T01-IM6-1<br>22 | P-00383<br>15-T01-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51584201           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.763-415:NCOA4_c.2137-51:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                              | NA                           |
| EVT-P-00607<br>65-T02-IM7-1<br>23 | P-00607<br>65-T02-IM7 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51584168           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.2137-188:RET_c.763-448NCOA4dup   Protein Fusion: out of frame {TIMM23B:RET}                                                              | NA                           |
| EVT-P-00430<br>77-T01-IM6-1<br>24 | P-00430<br>77-T01-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51583547           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+608:NCOA4_c.2137-452:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                             | NA                           |
| EVT-P-00057<br>48-T01-IM5-1<br>25 | P-00057<br>48-T01-IM5 |            | NCOA4-RET   | NCOA4        | in-frame     | three_prime | 51584356           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.763-260:NCOA4_c.2137-707:RETdup   Protein fusion: in frame (NCOA4-RET)                                                                  | NA                           |
| EVT-P-00479<br>91-T02-IM6-1<br>26 | P-00479<br>91-T02-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51583476           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.762+537:NCOA4_c.2137-494:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                             | NA                           |
| EVT-P-00488<br>05-T01-IM6-1<br>27 | P-00488<br>05-T01-IM6 |            | NCOA4-RET   | NCOA4        | unknown      | three_prime | 51584704           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.851:NCOA4_c.2137-807:RETdup   Protein Fusion: mid-exon {NCOA4:RET}                                                                       | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0049600-T01-IM6-128 | P-0049600-T01-IM6 |            | NCOA4-RET   | NCOA4        | unknown      | three_prime | 51586611           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.1889:NCOA4_c.2137-419:RETdup   Protein Fusion: mid-exon {NCOA4:RET}                                                                                  | NA                           |
| EVT-P-0056870-T01-IM6-129 | P-0056870-T01-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51582404           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion: c.618+132:NCOA4_c.2136+153:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                                         | NA                           |
| EVT-P-0052409-T01-IM6-130 | P-0052409-T01-IM6 |            | NCOA4-RET   | NCOA4        | out-of-frame | three_prime | 51582486           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | NCOA4 (NM_001145260) - RET (NM_020975) rearrangement: c.618+214:NCOA4_c.2137-341:RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                                  | NA                           |
| EVT-P-0004094-T02-IM6-131 | P-0004094-T02-IM6 |            | RELCH-RET   | RELCH        | in-frame     | three_prime | 43610279           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | KIAA1468 (NM_020854) - RET (NM_020975) fusion (KIAA1468 exons 1-10 fused with RET exons 12-20): t(10;18)(q11.21;q21.33)(chr10:g.43610279::chr18:g.59909399)   Protein Fusion: in frame {KIAA1468:RET} | NA                           |
| EVT-P-0015588-T01-IM6-132 | P-0015588-T01-IM6 |            | RET-CDC186  | CCDC186      | in-frame     | three_prime | 43610308           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | C10orf118 (NM_018017) - RET (NM_020975) Rearrangement : c.1326+142:C10orf118_c.2136+124:RETinv   Protein fusion: in frame (C10orf118-RET)                                                             | NA                           |
| EVT-P-0020974-T01-IM6-133 | P-0020974-T01-IM6 |            | RET-CDC6    | CCDC6        | in-frame     | three_prime | 43611184           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12): c.303+2543:CCDC6_c.2137-848:RETinv   Protein Fusion: in frame {CCDC6:RET}                                             | NA                           |
| EVT-P-0001745-T02-IM5-134 | P-0001745-T02-IM5 |            | RET-CDC6    | CCDC6        | in-frame     | three_prime | 43611245           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - (RET (NM_020975) rearrangement: c.304-9458:CCDC6_c.2137-787:RETinv   Protein fusion: in frame (CCDC6-RET)                                                                         | NA                           |
| EVT-P-0017365-T02-IM6-135 | P-0017365-T02-IM6 |            | RET-CDC6    | CCDC6        | in-frame     | three_prime | 43611464           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv   Protein Fusion: in frame {CCDC6:RET}                                        | NA                           |
| EVT-P-0001898-T01-IM3-136 | P-0001898-T01-IM3 |            | RET-CDC6    | CCDC6        | in-frame     | three_prime | 43611699           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) inversion: c.304-12339_c.2137-333inv5   Protein fusion: in frame (CCDC6-RET)                                                                                      | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0029602-T01-IM6-137 | P-0029602-T01-IM6 |            | RET-C CDC6  | CCDC6        | in-frame     | three_prime | 43611160           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-5768:CCDC6_c.2137-872:RETinv   Protein Fusion: in frame {CCDC6:RET}                                                 | NA                           |
| EVT-P-0004916-T01-IM5-138 | P-0004916-T01-IM5 |            | RET-C CDC6  | CCDC6        | in-frame     | three_prime | 43610353           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6(NM_005436)-RET(NM_020975) Fusion(CCDC6 exon 2 fused to RET exon11) : c.304-1153:CCDC6_c.2136+169:RETinv   Protein fusion: in frame (CCDC6-RET)                  | NA                           |
| EVT-P-0000425-T01-IM3-139 | P-0000425-T01-IM3 |            | RET-C CDC6  | CCDC6        | in-frame     | five_prime  | 43610450           | 11              | 712                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020630) - CCDC6 (NM_005436) Inversion (c.2136+266_c.303+19630inv)   Protein fusion: in frame (RET-CCDC6)                                                      | NA                           |
| EVT-P-0031330-T01-IM6-140 | P-0031330-T01-IM6 |            | RET-C CDC6  | CCDC6        | in-frame     | three_prime | 43610353           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20) : c.303+3703:CCDC6_c.2136+169:RETinv   Protein Fusion: in frame {CCDC6:RET}        | NA                           |
| EVT-P-0030990-T03-IM6-141 | P-0030990-T03-IM6 |            | RET-C CDC6  | CCDC6        | in-frame     | three_prime | 43611881           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exon 12-20): c.303+10940:CCDC6_c.2137-151:RETinv   Protein Fusion: in frame {CCDC6:RET}         | NA                           |
| EVT-P-0005168-T01-IM5-142 | P-0005168-T01-IM5 |            | RET-C CDC6  | CCDC6        | out-of-frame | three_prime | 43608797           | 10              | 587                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) Fusion ( CCDC6 Exon1 fused to RET Exon10 ) : c.304-11493:CCDC6_c.1760-207:RETinv   Protein fusion: out of frame (CCDC6-RET)       | NA                           |
| EVT-P-0022380-T01-IM6-143 | P-0022380-T01-IM6 |            | RET-C CDC6  | CCDC6        | in-frame     | three_prime | 43610659           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20) : c.303+21810:CCDC6_c.2136+475:RETinv   Protein Fusion: in frame {CCDC6:RET}       | NA                           |
| EVT-P-0022380-T02-IM6-144 | P-0022380-T02-IM6 |            | RET-C CDC6  | CCDC6        | in-frame     | three_prime | 43610659           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET(NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+21810:CCDC6_c.2136+475: c.2136+475inv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-0029315-T02-IM6-145 | P-0029315-T02-IM6 |            | RET-C CDC6  | CCDC6        | in-frame     | three_prime | 43610631           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.304-20462:CCDC6_c.2136+447:RETinv   Protein Fusion: in frame {CCDC6:RET}        | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                            | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00293<br>15-T01-IM6-146 | P-00293<br>15-T01-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43610631           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20: c.304-20462:CCDC6_c.2136+447:RETinv   Protein Fusion: in frame {CCDC6:RET})    | NA                           |
| EVT-P-00008<br>89-T01-IM3-147 | P-00008<br>89-T01-IM3 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | five_prime  | 43611994           | 12              | 713                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020630) - CCDC6 (NM_005436) Reciprocal Inversion: c.2137-38_c.2137-38inv   Protein fusion: in frame (RET-CCDC6)                                           | NA                           |
| EVT-P-00073<br>22-T02-IM6-148 | P-00073<br>22-T02-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43610268           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.454-1512:CCDC6_c.2136+84:RETinv   Protein Fusion: in frame {CCDC6:RET})                                             | NA                           |
| EVT-P-00279<br>53-T01-IM6-149 | P-00279<br>53-T01-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611278           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12-20): c.303+20901:CCDC6_c.2137-754:RETinv   Protein Fusion: in frame {CCDC6:RET})   | NA                           |
| EVT-P-00085<br>04-T03-IM6-150 | P-00085<br>04-T03-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43610583           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exons 1-8 fused to RET exons 12-20): c.1230+499:CCDC6_c.2136+399:RETinv   Protein Fusion: in frame {CCDC6:RET}) | NA                           |
| EVT-P-00089<br>55-T03-IM5-151 | P-00089<br>55-T03-IM5 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611019           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+26028:CCDC6_c.2136+835:RETinv   Protein fusion: in frame (CCDC6-RET)                                            | NA                           |
| EVT-P-00130<br>23-T01-IM5-152 | P-00130<br>23-T01-IM5 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611810           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.304-17904:CCDC6_c.2137-222:RETinv   Protein fusion: in frame (CCDC6-RET)                                            | NA                           |
| EVT-P-00133<br>94-T01-IM5-153 | P-00133<br>94-T01-IM5 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43610232           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion: c.303+18884:CCDC6_c.2136+48:RETinv   Protein fusion: in frame (CCDC6-RET)                                             | NA                           |
| EVT-P-00226<br>07-T01-IM6-154 | P-00226<br>07-T01-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611212           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon 1 fused to RET exons 12 - 20): c.304-414:CCDC6_c.2137-820:RETinv   Protein Fusion: in frame {CCDC6:RET})   | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains     | source_annotation_text                                                                                                                                         | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00226<br>42-T01-IM6-155 | P-00226<br>42-T01-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611768           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                       | CCDC6(NM_005436) - RET(NM_020975) Fusion (CCDC6 exon1 fused to RET exon12) : c.304-14214:CCDC6_c.2137-273:RETinv   Protein Fusion: in frame {CCDC6:RET}        | NA                           |
| EVT-P-00226<br>42-T02-IM6-156 | P-00226<br>42-T02-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611768           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                       | CCDC6 (NM_005436) - RET (NM_020975) Fusion (CCDC6 exon1 fused to RET exons 12-20) : c.304-14214:CCDC6_c.2137-273:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-00173<br>65-T03-IM6-157 | P-00173<br>65-T03-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611464           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                       | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 fused to RET exons 12-20): c.303+25501:CCDC6_c.2137-568:RETinv   Protein Fusion: in frame {CCDC6:RET}  | NA                           |
| EVT-P-00253<br>49-T03-IM6-158 | P-00253<br>49-T03-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611986           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                       | CCDC6 (NM_005436) - RET (NM_020975) fusion ( CCDC6 exon 1 fused to RET exons 12-20): c.304-10038:CCDC6_c.2137-46:RETinv   Protein Fusion: in frame {CCDC6:RET} | NA                           |
| EVT-P-00495<br>88-T01-IM6-159 | P-00495<br>88-T01-IM6 |            | RET-C<br>CDC6 | CCDC6        | out-of-frame | five_prime  | 43609437           | 10              | 627                         | True                   | lost          |                       | Protein kinase domain |                       | RET (NM_020975) - CCDC6 (NM_005436) rearrangement: c.1879+314:RET_c.304-15222:CCDC6inv   Protein Fusion: out of frame {RET:CCDC6}                              | NA                           |
| EVT-P-00154<br>35-T01-IM6-160 | P-00154<br>35-T01-IM6 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43610969           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                       | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 fused to RET exon 12): c.304-21366:CCDC6_c.2136+785:RETinv   Protein fusion: in frame (CCDC6-RET)      | NA                           |
| EVT-P-00055<br>56-T01-IM5-161 | P-00055<br>56-T01-IM5 |            | RET-C<br>CDC6 | CCDC6        | in-frame     | three_prime | 43611828           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                       | CCDC6 (NM_005436) - RET (NM_020975) fusion (CCDC6 exon1 with RET exons 12-20) - c.303+21888:CCDC6_c.2137-204:RETinv   Protein fusion: in frame (CCDC6-RET)     | NA                           |
| EVT-P-00635<br>02-T01-IM7-162 | P-00635<br>02-T01-IM7 |            | RET-CTNNA1    | CTNNA1       | unknown      | five_prime  | 43622038           | 19              | 1019                        | False                  | retained      | Protein kinase domain |                       |                       | RET (NM_020975) - CTNNA1 (NM_001903) fusion: t(5;10)(q31.2;q11.21)(chr5:g.138223827::chr10:g.43622038)   Protein Fusion: mid-exon {RET:CTNNA1}                 | NA                           |
| EVT-P-00638<br>02-T01-IM7-163 | P-00638<br>02-T01-IM7 |            | RET-CTNNA3    | CTNNA3       | out-of-frame | five_prime  | 43617566           | 16              | 934                         | True                   | disrupted     |                       |                       | Protein kinase domain | RET (NM_020975) - CTNNA3 (NM_013266) rearrangement: c.2801+102:RET_c.1375-34298:CTNNA3inv   Protein Fusion: out of frame {RET:CTNNA3}                          | NA                           |

| event_id                  | sample_id         | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0030396-T02-IM6-164 | P-0030396-T02-IM6 |            | RET-DOCK1   | DOCK1        | unknown      | three_prime | 43608320           | 9               | 556                         | False                  | retained      | Protein kinase domain |                       |                   | DOCK1 (NM_001380) - RET (NM_020975) Rearrangement: c.2515+295:RET_c.1668:RETdup   Protein Fusion: mid-exon {DOCK1:RET}                                | NA                           |
| EVT-P-0013886-T01-IM5-165 | P-0013886-T01-IM5 |            | RET-EML4    | EML4         | unknown      | three_prime | 43610067           | 11              | 673                         | False                  | retained      | Protein kinase domain |                       |                   | EML4 (NM_019063) - RET (NM_020975) rearrangement: t(2;10)(p21;q11.21)(chr2:g.42550559::chr10:g.43610067)   Protein fusion: mid-exon (EML4-RET)        | NA                           |
| EVT-P-0014808-T03-IM6-166 | P-0014808-T03-IM6 |            | RET-ERCC6   | ERCC6        | out-of-frame | five_prime  | 43611216           | 12              | 713                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6 inv   Protein Fusion: out of frame {RET:ERCC6}                    | NA                           |
| EVT-P-0014808-T02-IM6-167 | P-0014808-T02-IM6 |            | RET-ERCC6   | ERCC6        | out-of-frame | five_prime  | 43611216           | 12              | 713                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6 inv   Protein Fusion: out of frame {RET:ERCC6}                    | NA                           |
| EVT-P-0014808-T01-IM6-168 | P-0014808-T01-IM6 |            | RET-ERCC6   | ERCC6        | out-of-frame | five_prime  | 43611216           | 12              | 713                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - ERCC6 (NM_000124) rearrangement: c.2137-816:RET_c.1397+6822:ERCC6 inv   Protein fusion: out of frame (RET-ERCC6)                    | NA                           |
| EVT-P-0045809-T02-IM6-169 | P-0045809-T02-IM6 |            | RET-FBXL7   | FBXL7        | in-frame     | five_prime  | 43622269           | 19              | 1063                        | True                   | retained      | Protein kinase domain |                       |                   | RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622269)   Protein Fusion: in frame {RET:FBXL7}           | NA                           |
| EVT-P-0045809-T06-IM6-170 | P-0045809-T06-IM6 |            | RET-FBXL7   | FBXL7        | in-frame     | five_prime  | 43622323           | 19              | 1063                        | True                   | retained      | Protein kinase domain |                       |                   | RET (NM_020975) - FBXL7 (NM_012304) fusion: t(5;10)(p15.1;q11.21)(chr5:g.15556734::chr10:g.43622323)   Protein Fusion: in frame {RET:FBXL7}           | NA                           |
| EVT-P-0025215-T01-IM6-171 | P-0025215-T01-IM6 |            | RET-GEMIN5  | GEMIN5       | in-frame     | three_prime | 43611067           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | GEMIN5 (NM_015465) - RET (NM_020975) rearrangement: t(5;10)(q33.2;q11.21)(chr5:g.154274993::chr10:g.43611067)   Protein Fusion: in frame {GEMIN5:RET} | NA                           |
| EVT-P-0027417-T01-IM6-173 | P-0027417-T01-IM6 |            | RET-KCNMA1  | KCNMA1       | out-of-frame | three_prime | 43608158           | 9               | 550                         | True                   | retained      | Protein kinase domain |                       |                   | KCNMA1 (NM_001161352) - RET (NM_020975) rearrangement: c.378+15627:KCNMA1_c.1649-143:RET   Protein Fusion: out of frame {KCNMA1:RET}                  | NA                           |
| EVT-P-0043704-T01-IM6-174 | P-0043704-T01-IM6 |            | RET-KIF5B   | KIF5B        | in-frame     | five_prime  | 32306364           | 1               | 1                           | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - KIF5B (NM_004521) rearrangement: c.2136+147:RET_c.2545-77:KIF5B inv   Protein Fusion: in frame {RET:KIF5B}                          | NA                           |

| event_id                      | sample_id             | patient_id | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                              | source_site2_effect_on_frame |
|-------------------------------|-----------------------|------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00062<br>26-T02-IM5-178 | P-00062<br>26-T02-IM5 |            | RET-N<br>COA4  | NCOA4        | in-frame     | thre_prime  | 43611893           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion : c.762+254:NCOA4_c.2137-139:RETdup   Protein fusion: in frame (NCOA4-RET)                                            | NA                           |
| EVT-P-00011<br>51-T01-IM3-179 | P-00011<br>51-T01-IM3 |            | RET-N<br>COA4  | NCOA4        | in-frame     | thre_prime  | 43611645           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2137-387_c.1840-75dup   Protein fusion: in frame (NCOA4-RET)                                                    | NA                           |
| EVT-P-00095<br>92-T01-IM5-180 | P-00095<br>92-T01-IM5 |            | RET-N<br>COA4  | NCOA4        | unknown      | thre_prime  | 43611143           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-9 with RET exons 12=20: c.872:NCOA4_c.2136+621:RETdup   Protein fusion: mid-exon (NCOA4-RET)           | NA                           |
| EVT-P-00011<br>55-T01-IM3-181 | P-00011<br>55-T01-IM3 |            | RET-N<br>COA4  | NCOA4        | in-frame     | thre_prime  | 43610941           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | RET (NM_020630) - NCOA4 (NM_005437) Duplication : c.2136+757_c.715-617dup   Protein fusion: in frame (NCOA4-RET)                                                    | NA                           |
| EVT-P-00228<br>18-T01-IM6-182 | P-00228<br>18-T01-IM6 |            | RET-N<br>COA4  | NCOA4        | unknown      | thre_prime  | 43611746           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup   Protein Fusion: mid-exon {NCOA4:RET} | NA                           |
| EVT-P-00021<br>45-T01-IM3-183 | P-00021<br>45-T01-IM3 |            | RET-P<br>DCD10 | PDCD10       | out-of-frame | five_prime  | 43609394           | 10              | 627                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - PDCD10 (NM_145860) translocation: t(10,3)(q11.21;q26.1)(chr10:g.43609394::chr3:g.167419184)   Protein fusion: out of frame (RET-PDCD10)           | NA                           |
| EVT-P-00018<br>98-T01-IM3-184 | P-00018<br>98-T01-IM3 |            | RET-R<br>ASSF4 | RASSF4       | in-frame     | five_prime  | 43611749           | 12              | 713                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - RASSF4 (NM_032023) deletion: c.2137-283_c.138+357del   Protein fusion: in frame (RET-RASSF4)                                                      | NA                           |
| EVT-P-00303<br>96-T02-IM6-185 | P-00303<br>96-T02-IM6 |            | RET-R<br>BPMS  | RBPMS        | unknown      | thre_prime  | 43610018           | 11              | 657                         | False                  | retained      | Protein kinase domain |                       |                   | RBPMS (NM_001008712) - RET (NM_020975) Rearrangement : t(8,10)(p21.1,q11.21)(chr8:g.30412656::chr10:g.43610018)   Protein Fusion: mid-exon {RBPMS:RET}              | NA                           |
| EVT-P-00292<br>22-T02-IM6-212 | P-00292<br>22-T02-IM6 |            | RET-R<br>UFY1  | RUFY1        | unknown      | thre_prime  | 43610037           | 11              | 663                         | False                  | retained      | Protein kinase domain |                       |                   | RUFY1 (NM_025158) - RET (NM_020975) Rearrangement : t(5,10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037)   Protein Fusion: mid-exon {RUFY1:RET}                | NA                           |
| EVT-P-00153<br>10-T02-IM6-213 | P-00153<br>10-T02-IM6 |            | RET-R<br>UFY3  | RUFY3        | out-of-frame | thre_prime  | 43610427           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655316::chr10:g.43610427)   Protein fusion: out of frame (RUFY3-RET)           | NA                           |

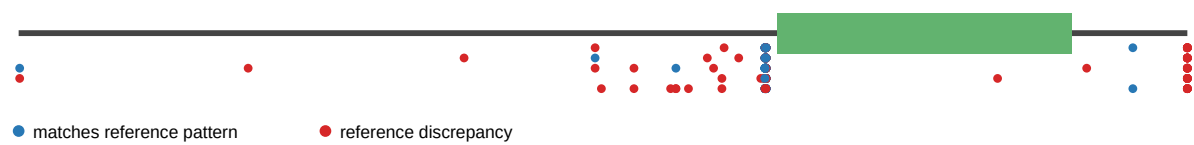
| event_id                          | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                              | source_site2_effect_on_frame |
|-----------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00153<br>10-T04-IM6-2<br>14 | P-00153<br>10-T04-IM6 |            | RET-RUFY3   | RUFY3        | unknown      | three_prime | 43610427           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | RUFY3 (NM_001037442) - RET (NM_020975) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.71655251::chr10:g.43610427)   Protein Fusion: mid-exon {RUFY3:RET}                               | NA                           |
| EVT-P-00256<br>09-T01-IM6-2<br>15 | P-00256<br>09-T01-IM6 |            | RET-SHROOM3 | SHROOM3      | in-frame     | three_prime | 43611545           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | SHROOM3 (NM_020859) - RET (NM_020975) rearrangement: t(4;10)(q21.1;q11.21)(chr4:g.77664882::chr10:g.43611545)   Protein Fusion: in frame {SHROOM3:RET}                              | NA                           |
| EVT-P-00535<br>30-T01-IM6-2<br>16 | P-00535<br>30-T01-IM6 |            | RET-SLC4A4  | SLC4A4       | out-of-frame | five_prime  | 43610439           | 11              | 712                         | True                   | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - SLC4A4 (NM_001134742) rearrangement: t(4;10)(q13.3;q11.21)(chr4:g.72412310::chr10:g.43610439)   Protein Fusion: out of frame {RET:SLC4A4}                         | NA                           |
| EVT-P-00021<br>45-T01-IM3-2<br>17 | P-00021<br>45-T01-IM3 |            | RET-TFG     | TFG          | in-frame     | three_prime | 43609437           | 10              | 627                         | True                   | retained      | Protein kinase domain |                       |                   | RET (NM_020975) - TFG (NM_001195478) translocation: t(10;3)(q11.21;q12.2)(chr10:g.43609437::chr3:g.100450751)   Protein fusion: in frame (TFG-RET)                                  | NA                           |
| EVT-P-00219<br>80-T01-IM6-2<br>18 | P-00219<br>80-T01-IM6 |            | RET-TIMM23B | TIMM23B      | out-of-frame | three_prime | 43610550           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-8 fused to RET exons 12-20): c.763-591:NCOA4_c.2136+366:RETdup   Protein Fusion: out of frame {TIMM23B:RET}            | NA                           |
| EVT-P-00177<br>16-T01-IM6-2<br>19 | P-00177<br>16-T01-IM6 |            | RET-TIMM23B | TIMM23B      | out-of-frame | three_prime | 43611172           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | NCOA4 (NM_001145260) - RET (NM_020975) fusion (NCOA4 exons 1-7 fused with RET exons 12-20): c.618+42:NCOA4_c.2137-860:RETdup   Protein Fusion: out of frame {TIMM23B:RET}           | NA                           |
| EVT-P-00255<br>31-T01-IM6-2<br>20 | P-00255<br>31-T01-IM6 |            | RET-TIMM23B | TIMM23B      | out-of-frame | three_prime | 43611375           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | NCOA4 (NM_001145260) RET (NM_020975) fusion: c.619-108NCOA4_c.2137-657RETdup   Protein Fusion: out of frame {TIMM23B:RET}                                                           | NA                           |
| EVT-P-00347<br>04-T01-IM6-2<br>22 | P-00347<br>04-T01-IM6 |            | RET-TRIM24  | TRIM24       | in-frame     | three_prime | 43611827           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | TRIM24 (NM_015905) - RET (NM_020975) Fusion (TRIM24 exon 9 fused with RET exon12) : t(7;10)(q34;q11.21)(chr7:g.138243011::chr10:g.43611827)   Protein Fusion: in frame {TRIM24:RET} | NA                           |
| EVT-P-00467<br>90-T01-IM6-2<br>23 | P-00467<br>90-T01-IM6 |            | RET-UXS1    | UXS1         | unknown      | five_prime  | 43600430           | 4               | 219                         | False                  | lost          |                       | Protein kinase domain |                   | RET (NM_020975) - UXS1 (NM_001253875) rearrangement: t(2;10)(q12.2;q11.21)(chr2:g.106728260::chr10:g.43600430)   Protein Fusion: mid-exon {RET:UXS1}                                | NA                           |

| event_id                     | sample_id             | patient_id | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | domain_status | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                                                  | source_site2_effect_on_frame |
|------------------------------|-----------------------|------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|---------------|-----------------------|-----------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-00292<br>22-T03-IM6-24 | P-00292<br>22-T03-IM6 |            | RUFY1-RET   | RUFY1        | unknown      | thre_prime  | 179025890          | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | RUFY1 (NM_025158) - RET (NM_020975) fusion: t(5;10)(q35.3;q11.21)(chr5:g.179025890::chr10:g.43610037)   Protein Fusion: mid-exon {RUFY1:RET}                                                            | NA                           |
| EVT-P-00313<br>40-T02-IM6-25 | P-00313<br>40-T02-IM6 |            | RUFY2-RET   | RUFY2        | in-frame     | thre_prime  | 70141796           | 20              | 1115                        | True                   | lost          |                       | Protein kinase domain |                   | RUFY2 (NM_017987) - RET (NM_020975) fusion: c.1045-640:RUFY2_c.2137-473:RETinv   Protein Fusion: in frame {RUFY2:RET}                                                                                   | NA                           |
| EVT-P-00266<br>14-T01-IM6-27 | P-00266<br>14-T01-IM6 |            | SPECC1L-RET | SPECC1L      | in-frame     | thre_prime  | 43611575           | 12              | 713                         | True                   | retained      | Protein kinase domain |                       |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-10 fused to RET exons 12-20): t(10;22)(10q11.21;22q11.23)(chr10:g.43611575::chr22:g.24740413)   Protein Fusion: in frame {SPECC1L:RET} | NA                           |
| EVT-P-00364<br>00-T06-IM6-28 | P-00364<br>00-T06-IM6 |            | SPECC1L-RET | SPECC1L      | in-frame     | thre_prime  | 43610353           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion: t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056)   Protein Fusion: in frame {SPECC1L:RET}                                                   | NA                           |
| EVT-P-00364<br>00-T04-IM6-29 | P-00364<br>00-T04-IM6 |            | SPECC1L-RET | SPECC1L      | in-frame     | thre_prime  | 43610353           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056)   Protein Fusion: in frame {SPECC1L:RET}      | NA                           |
| EVT-P-00364<br>00-T03-IM6-30 | P-00364<br>00-T03-IM6 |            | SPECC1L-RET | SPECC1L      | in-frame     | thre_prime  | 43610353           | 11              | 712                         | True                   | retained      | Protein kinase domain |                       |                   | SPECC1L (NM_001145468) - RET (NM_020975) fusion (SPECC1L exons 1-9 fused to RET exons 12-20): t(10;22)(q11.21;q11.23)(chr10:g.43610353::chr22:g.24739056)   Protein Fusion: in frame {SPECC1L:RET}      | NA                           |

# Figures

## domain retention outliers

RET domain-retention track



## reference comparison

Reference vs msk\_impact\_50k\_2026

